

Notes: Xiouros & Zapatero (JFE, 2024)

Disagreement, information quality and asset prices

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1 Big picture

- **Question.** How do disagreement among investors and time variation in information quality affect stock prices, bond prices, risk premia, and real yields?
- **Core mechanism.** Investors receive a common informative signal about the latent growth rate of the economy. When the quality of this common signal deteriorates, investors rely more on idiosyncratic or “tribal” signals. This raises disagreement and forecast dispersion.
- **Main theoretical innovation.** The paper solves a pure exchange economy with a continuum of agents, heterogeneous beliefs, time-varying information quality, and external habit preferences. The continuum of types makes the equilibrium stationary.
- **Main empirical calibration.** The belief process is estimated with Survey of Professional Forecasters data on GDP growth forecasts and aggregate consumption data. Asset pricing moments are then evaluated using stock market, T-bill, and real yield data.
- **Main stock-market result.** Constant disagreement has little quantitative impact on stock prices. The important force is *variation* in disagreement, generated by shocks to information quality.
- **Main volatility result.** In the benchmark model, time-varying information quality plus disagreement generates a substantial part of observed stock price volatility. Removing disagreement sharply lowers price-dividend volatility.
- **Main equity-premium result.** Information quality is procyclical in the model. Bad aggregate shocks lower information quality, raise disagreement, and lower stock prices. This makes information-quality risk systematic and raises the equity premium.
- **Main bond-market result.** Market forecast dispersion affects the risk-free rate and helps generate an upward-sloping real yield curve. The model-implied long real yields are positively correlated with real yields in the data, especially at long maturities.
- **Main methodological message.** A continuum of belief types changes the implications of heterogeneous-belief models. In two-agent models, wealth redistribution creates persistent nonstationarity. With a continuum, the law of large numbers stabilizes the wealth-weighted distribution of beliefs.
- **Economic takeaway.** Disagreement by itself is not enough. Disagreement becomes quantitatively important when information quality moves over time and when these movements are tied to aggregate shocks.

2 Data and measurement

2.1 Core data sources

- **Aggregate consumption.** The macro state is disciplined with real per-capita consumption of nondurables and services. Consumption growth is used to estimate the latent growth-rate process and to anchor the economy’s aggregate endowment.
- **Survey expectations.** The disagreement process is disciplined with Survey of Professional Forecasters quarter-ahead real GDP growth forecasts. The paper uses both the cross-sectional mean forecast and the cross-sectional dispersion of forecasts.
- **Stock market data.** Stock return moments come from the CRSP value-weighted market return, including dividends. The price-dividend ratio is formed from market prices and aggregate dividends.
- **Short-rate data.** The risk-free rate is measured using the three-month T-bill rate, adjusted for expected inflation when the paper studies real rates.
- **Real-yield data.** The real term-structure tests use real yields, especially TIPS-based long-maturity real yields, to evaluate whether the model-implied real yield curve moves with the data.

Interpretation: The empirical design is not a standard treatment-control design. The paper is a structural/quantitative asset-pricing exercise: forecast data discipline the belief process, consumption data discipline the endowment process, and asset-price data are used to evaluate the model’s pricing implications.

2.2 Measured objects and model analogues

Data object	Model object	Role in the paper
Consumption growth	$g_{t+1} = \log(C_{t+1}/C_t)$	Pins down aggregate endowment dynamics and the latent growth process.
Mean forecast	$\bar{\mu}_t$ and market expectation $\tilde{\mu}_t$	Measures expected growth and tests whether growth expectations alone can explain valuations.
Forecast dispersion	$\bar{\nu}_t$ and market dispersion $\tilde{\nu}_t$	Main empirical proxy for disagreement generated by low information quality.
Information quality	ρ_t and persistent state χ_t	Latent driver of disagreement; lower quality raises reliance on individual signals.
Risk-free rate	Model-implied short real rate	Tests whether dispersion predicts future short rates and real yields.
Stock valuation	Price-dividend ratio	Tests whether information-quality risk and disagreement generate excess volatility.
Real yields	Model-implied real term structure	Evaluates whether the disagreement channel generates upward-sloping real bond premia.

2.3 Why professional forecasts are central

- The paper needs a direct measure of disagreement, rather than inferring disagreement only from asset prices.
- Professional forecasters provide repeated cross-sections of beliefs about near-term macro growth.

- The cross-sectional dispersion of forecasts is used to identify the magnitude, volatility, and persistence of disagreement.
- The mean forecast is used to discipline learning about expected growth.
- The model uses forecast moments first, then asks whether the implied belief process can help price stocks and bonds.

Economic message: This is the key empirical strength of the paper. The disagreement process is not reverse-engineered from asset prices; it is estimated from survey beliefs and then taken to asset pricing.

2.4 Important measurement caveats

- Professional forecasters are not literally the marginal investors in all asset markets.
- GDP forecasts are not the same as consumption-growth forecasts, though they are closely related macro expectations.
- Forecast dispersion is an observed proxy for a richer latent belief distribution.
- Market forecast dispersion is wealth weighted in the model, while survey dispersion is usually unweighted.
- TIPS data are available only for a relatively recent sample, so real-yield tests are shorter-sample validation exercises.

Interpretation: The paper’s empirical strategy is credible as a disciplined quantitative exercise, but the mapping from survey forecasters to market investors is an approximation.

3 Model structure and equations

3.1 State variables

The model’s pricing state can be summarized by:

- latent mean consumption growth μ_t ;
- market expectation $\tilde{\mu}_t$;
- uncertainty about the latent mean v_t ;
- market forecast dispersion $\tilde{v}_t$;
- information-quality state χ_t and signal quality ρ_t ;
- habit/risk-aversion state ω_t ;
- speculative activity ξ_t , which summarizes belief-heterogeneity effects on state prices.

Interpretation: The paper separates belief states from preference states. This allows the authors to ask whether disagreement itself matters, or whether the model is simply using habit to fit asset prices.

3.2 Aggregate endowment block

$$g_{t+1} = \log \left(\frac{C_{t+1}}{C_t} \right), \quad (1)$$

$$g_{t+1} \mid \mu_t \sim N(\mu_t, \sigma_c^2), \quad (2)$$

$$\mu_{t+1} = \phi_\mu \mu_t + (1 - \phi_\mu) \mu_c + \sigma_\mu \varepsilon_{t+1}^\mu. \quad (3)$$

Interpretation: The economy has long-run growth news because agents learn about μ_t . However, the paper shows that this expected-growth channel alone is not strong enough to explain stock-market volatility.

3.3 Information and disagreement block

The common signal is

$$s_t^m = \rho_t \varepsilon_t^\mu + \sqrt{1 - \rho_t^2} \varepsilon_t^m. \quad (4)$$

The true individual signal is

$$s_t^i = \eta s_t^m + \sqrt{1 - \eta^2} \varepsilon_t^i. \quad (5)$$

Agent i believes her signal has loading

$$s_t^i = \zeta_t \varepsilon_t^\mu + \sqrt{1 - \zeta_t^2} \varepsilon_t^i, \quad \zeta_t = \frac{\eta}{\rho_t}. \quad (6)$$

Interpretation: When ρ_t is low, common information is weak. The perceived loading of individual signals rises, so agents rely more on idiosyncratic interpretations and disagreement rises.

3.4 Information-quality dynamics

$$\rho_t = \eta + (1 - \eta) \Phi(\chi_t + \sigma_\rho \varepsilon_t^\rho), \quad (7)$$

$$\chi_t = \phi_\chi \chi_{t-1} + (1 - \phi_\chi) \mu_\chi + \beta_\chi (g_t - \tilde{\mu}_{t-1}) + \sigma_\chi \varepsilon_t^\chi. \quad (8)$$

Interpretation: The parameter β_χ makes information quality respond to growth surprises. This creates systematic information-quality risk: bad macro news can lower information quality, raise disagreement, and depress asset prices.

3.5 Preferences and the stochastic discount factor

Agents have external-habit utility:

$$U_i = E_i \sum_{t=0}^{\infty} \beta^t \frac{(C_t^i - H_t)^{1-\gamma}}{1-\gamma}. \quad (9)$$

The marginal utility component from habit gives a homogeneous-belief SDF, while disagreement adds a belief-distortion component:

$$M_{t+1} = N_{t+1} \times \text{belief-distortion term}. \quad (10)$$

Interpretation: Habit is needed to generate a high price of risk. Belief heterogeneity changes the distribution of state prices and creates additional volatility and term-structure effects.

3.6 Consumption-share reallocation

The surplus consumption share of agent i evolves according to

$$\alpha_{t+1}^i = \alpha_t^i \frac{\phi(g_{t+1}; \mu_t^i, \sigma_t^2)^{1/\gamma}}{\int_0^1 \alpha_t^j \phi(g_{t+1}; \mu_t^j, \sigma_t^2)^{1/\gamma} dj}. \quad (11)$$

Interpretation: Agents who assigned high probability to realized growth gain surplus consumption share. Asset prices therefore depend on the wealth-weighted belief distribution, not merely the average forecast.

4 Identification, calibration, and mechanism tests

4.1 What is identified from non-asset-price data

- Consumption moments identify the aggregate endowment process.
- Forecast mean moments identify the learning dynamics for expected growth.
- Forecast dispersion moments identify the strength and persistence of disagreement.
- The relation between realized growth surprises and dispersion helps identify the systematic component of information quality.

Interpretation: The belief process is identified from macro and survey data before the paper asks whether the model can match asset prices.

4.2 What is calibrated from asset prices

- The discount factor is chosen to match the average risk-free rate.
- Preference curvature and habit parameters help match risk-free-rate volatility, the stock-market Sharpe ratio, and price-dividend persistence.
- Dividend parameters are chosen to match stock dividend-growth moments.

Interpretation: The calibration gives the model the same basic asset-pricing scale as the data. The mechanism tests then ask which ingredient is responsible for volatility, premia, and real yields.

4.3 The nested-model identification logic

Model	Risk aversion	Information quality	Purpose
1	Time varying	Stochastic with disagreement	Full benchmark.
2	Constant	Stochastic with disagreement	Isolate information-quality/disagreement without habit variation.
3	Constant	Constant	Show constant disagreement is weak.
4	Time varying	Constant	Habit without stochastic information quality.
5	Time varying	Stochastic but no disagreement	Test whether information quality matters without belief dispersion.
6	Constant	Stochastic but no disagreement	Minimal disagreement-free benchmark.

Economic message: The paper's main identification of mechanisms comes from ablation. If stochastic information quality matters only when disagreement is present, then the effect is not just uncertainty; it is disagreement generated by bad information.

4.4 How the paper validates the mechanism

- **Stock-market validation:** the full model generates substantially higher price-dividend volatility than models without time-varying disagreement.
- **Equity-premium validation:** stochastic information quality makes stock volatility systematic because bad growth news lowers information quality and raises disagreement.

- **Short-rate validation:** forecast dispersion predicts future short rates in the data and in the model.
- **Real-yield validation:** the full model produces positive correlations with long real yields, whereas stripped-down models perform poorly.
- **Incomplete-market validation:** restricting trade on individual signals has small quantitative effects under the calibration.

Interpretation: These are not causal treatment effects. They are model-validation exercises showing that one structural mechanism reproduces multiple facts across surveys, stocks, short rates, and real bonds.

4.5 Main limitations

- Survey forecasters may not represent marginal investors.
- The model uses an external-habit preference block, so some asset-pricing success depends on a standard habit mechanism.
- Real-yield tests are constrained by the shorter TIPS sample.
- Complete markets are tractable but strong, even though the paper argues incomplete markets do not materially change the benchmark calibration.
- The model abstracts from institutions, leverage constraints, and intermediary balance sheets that may also affect disagreement and bond premia.

Interpretation: The paper is strongest as a unified quantitative mechanism, not as a clean causal experiment.

5 Figure and table section

5.1 Tables 1–2: disagreement process estimation and parameters

Read: Table 1:

- The model matches annual consumption growth mean, volatility, and autocorrelation.
- It matches the volatility and autocorrelation of the mean forecast and forecast dispersion from the Survey of Professional Forecasters.
- The mean forecast predicts growth, while changes in the mean forecast are only weakly correlated with realized growth.

Read: Table 2:

- The estimated disagreement parameter is $\eta = 0.247$.
- The persistent information-quality component is highly persistent, with $\phi_\chi = 0.965$.
- The information-quality process has both persistent and transitory components.

Economic message: The estimated belief process is empirically disciplined and produces persistent forecast dispersion, which is crucial for asset pricing.

Table 1

Disagreement model estimation. The real per-capita consumption growth series (g) is annual, was computed from the per-capita consumption of non-durable goods and services (NIPA), deflated by the annual CPI index (BEA), and cover the period from 1930 to 2017. The beliefs data, where $\bar{\mu}$ denotes the mean forecast and $\bar{\nu}$ the standard deviation of forecasts, were obtained from the Survey of Professional Forecasters, are quarterly, span the period from 1968Q3 to 2016Q4, and correspond to the real GDP quarter-ahead forecasts. The data standard errors (*s.e.*) were Newey and West (1987) estimated using 16 lags. The model statistics are the averages across 1,000 simulations of 400 quarters each (with a burn-in of 100 quarters). The model t-statistics ($t - stat.$) correspond to the differences between the model and the data statistics. For a variable x , $\mu(x)$ denotes the sample mean, $\sigma(x)$ the standard deviation, $ac_k(x)$ its k -lag autocorrelation and $corr(x, y)$ the correlation with variable y . Each model was estimated using the Simulated Method of Moments, matching all moments shown, except the time-series mean of the cross-section average forecast $\mu(\bar{\mu})$ because this is given by μ_c , the unconditional mean of g . The estimated parameters are shown in Table 2.

Statistic	Description	Data		Model	
		<i>est.</i>	<i>s.e.</i>	<i>avg.</i>	<i>t - st</i>
A. Annual (time-aggregated) consumption growth (data: 1930 – 2017)					
$\mu(g^a)$	Mean consumption growth	1.74	(0.27)	1.74	(0.01)
$\sigma(g^a)$	Volatility of consumption growth	2.70	(0.67)	2.70	(0.00)
$ac_1(g^a)$	Autocorrelation of consumption growth	0.48	(0.05)	0.50	(0.49)
B. Quarterly mean forecast, forecast dispersion and growth (data: 1968Q3 – 2016Q4)					
$\mu(\bar{\mu})$	Average of mean forecast	0.39	(0.05)	0.44	(0.90)
$\sigma(\bar{\mu})/\sigma(g)$	Standardized volatility of mean forecast	0.47	(0.03)	0.49	(0.51)
$ac_1(\bar{\mu})$	Autocorrelation of mean forecast	0.80	(0.03)	0.77	(0.92)
$\mu(\bar{\nu})/\sigma(g)$	Standardized average of forecast dispersion	0.27	(0.03)	0.28	(0.08)
$\sigma(\bar{\nu})/\sigma(g)$	Standardized volatility of forecast dispersion	0.15	(0.01)	0.15	(0.11)
$ac_1(\bar{\nu})$	Lag 1 autocorrelation of forecast dispersion	0.71	(0.05)	0.77	(1.28)
$ac_4(\bar{\nu})$	Lag 4 autocorrelation of forecast dispersion	0.58	(0.08)	0.52	(0.72)
$ac_8(\bar{\nu})$	Lag 8 autocorrelation of forecast dispersion	0.43	(0.05)	0.42	(0.34)
$ac_{12}(\bar{\nu})$	Lag 12 autocorrelation of forecast dispersion	0.32	(0.05)	0.34	(0.34)
$corr(\bar{\mu}_t, g_{t+1})$	Growth predictability by average forecast	0.54	(0.05)	0.51	(0.52)
$corr(\Delta\bar{\mu}, g)$	Correlation of average forecast changes with growth	0.13	(0.07)	0.16	(0.34)
$corr(\bar{\nu}_t, g_{t+1})$	Growth predictability by forecast dispersion	-0.06	(0.11)	-0.11	(0.41)
$corr(\Delta\bar{\nu}, g)$	Correlation of forecast dispersion changes with growth	-0.20	(0.07)	-0.17	(0.39)
$corr(\bar{\mu}, \bar{\nu})$	Correlation between forecast mean and dispersion	-0.28	(0.17)	-0.20	(0.48)

Table 2

Estimated model parameters - quarterly frequency.

Description	Parameter	Value
Unconditional mean of consumption growth	μ_c	0.436%
Volatility of consumption growth shocks	σ_c	0.754%
Volatility of shocks to consumption growth conditional mean	σ_μ	0.474%
Persistence of consumption growth conditional mean	φ_μ	0.773
Correlation between common and individual signals	η	0.247
Volatility of common signal informativeness idiosyncratic shocks	σ_ν	1.11
Mean of common signal informativeness factor	μ_ν	-0.366
Persistence of common signal informativeness factor	φ_ν	0.965
Sensitivity of common signal informativeness factor to growth shocks	β_ν	30.0
Volatility of common signal informativeness factor idiosyncratic shocks	σ_β	0.257%

Table 1: disagreement model estimation

Table 2: estimated parameters

5.2 Figures 1–2: static effects of disagreement and information quality

Read: Figure 1:

- The figure shows belief-related quantities as information quality/disagreement varies.
- Higher disagreement increases market forecast dispersion and speculative activity.
- Better information quality reduces uncertainty and dispersion.

Read: Figure 2:

- The risk-free rate rises with disagreement for empirically relevant risk-aversion parameters.
- The price of risk is driven more by habit/risk aversion than by disagreement alone.
- Static disagreement has visible effects on interest rates but limited independent effects on stock price volatility.

Economic message: Figures 1 and 2 motivate the quantitative results: disagreement is most powerful through forecast dispersion and the short-rate channel.

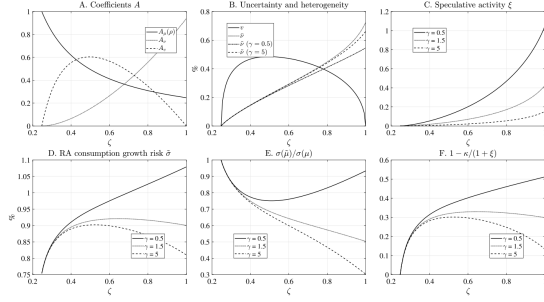


Figure 1: belief-related quantities

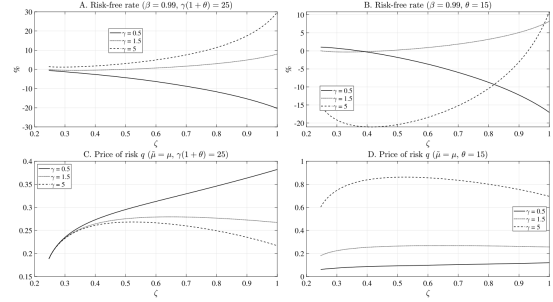


Figure 2: risk-free rate and price of risk

5.3 Tables 3–4: asset pricing moments and model configurations

Read: Table 3:

- Model 1, the full model, produces substantial price-dividend volatility and a realistic Sharpe ratio.
- Model 2 removes time-varying risk aversion but keeps stochastic information quality and disagreement; price-dividend volatility falls.
- Model 3 additionally makes information quality constant; price-dividend volatility collapses.
- CRRA preferences produce a poor market price of risk even when other dynamics remain similar.

Read: Table 4:

- Panel A lists common preference and dividend parameters.
- Panel B defines how Models 1–6 differ by risk aversion, information quality, and disagreement.

Economic message: The nested models reveal the interaction result: stochastic information quality matters for stocks only when it changes disagreement.

Table 3
Asset pricing statistics – Data and models with disagreement. The table shows a number of asset pricing moments for the data and four different model configurations. Model 1 refers to the full model, Model 2 refers to the model with constant risk aversion and Model 3 refers to the model with both constant risk aversion and constant information quality—see parameters in Tables 2 and 4. CRRA is as Model 2, but with CRRA preferences ($\theta = 0$), where we set $\beta = 0.9972$ to match the average pd to that of Model 2. For a variable x , $\mu(x)$ denotes the sample mean, $\sigma(x)$ the standard deviation, $ac_k(x)$ its k -lag autocorrelation and $corr(x, y)$ the correlation with variable y . The notation is as follows: pd denotes the log price-to-dividend ratio for the stock market dividend claim, r^f the log risk-free rate, r^m the log stock market return, SR_m the stock market Sharpe ratio using log returns, pc the log price-to-consumption ratio of the per-capita consumption claim, and r^c the log return of the consumption claim. The standard errors for the data estimates *s.e.* are computed using the Newey and West (1987) method with 16 lags. For each model we report the averages of 1,000 simulations of 400 quarters each (with a burn-in of 100 quarters). The t -statistics for the models $t - st$ refer to the hypotheses that the data estimates are equal to the model averages.

	Data		Model 1		Model 2		Model 3		CRRA	
	Data	s.e.	avg.	$t - st$	avg.	$t - st$	avg.	$t - st$	avg.	$t - st$
$\mu(pd)$	4.71	(0.09)	4.34	(4.10)	4.64	(0.77)	4.71	(0.10)	4.63	(0.83)
$\sigma(pd)$	0.44	(0.05)	0.31	(2.38)	0.16	(5.33)	0.02	(7.83)	0.15	(5.52)
$ac_1(pd)$	0.97	(0.01)	0.96	(0.88)	0.94	(2.22)	0.76	(14.17)	0.93	(2.36)
$\mu(r^f)$	0.14	(0.13)	0.16	(0.16)	0.16	(0.17)	0.79	(4.85)	1.73	(11.87)
$\sigma(r^f)$	0.84	(0.09)	0.85	(0.10)	0.85	(0.11)	0.61	(2.44)	0.96	(1.28)
$ac_1(r^f)$	0.65	(0.05)	0.65	(0.03)	0.65	(0.03)	0.76	(2.24)	0.66	(0.15)
$\mu(r^m)$	1.59	(0.47)	1.82	(0.50)	1.41	(0.37)	1.33	(0.55)	1.42	(0.36)
$\sigma(r^m)$	10.52	(1.55)	11.66	(0.74)	8.99	(0.99)	6.55	(2.56)	8.78	(1.12)
$ac_1(r^m)$	-0.03	(0.09)	0.01	(0.54)	0.02	(0.58)	0.03	(0.69)	0.02	(0.59)
$\mu(r^m - r^f)$	1.44	(0.49)	1.66	(0.44)	1.25	(0.39)	0.54	(1.84)	-0.31	(3.56)
$\sigma(r^m - r^f)$	10.53	(1.55)	11.61	(0.70)	8.94	(1.02)	6.52	(2.58)	8.73	(1.16)
$ac_1(r^m - r^f)$	-0.03	(0.09)	-0.00	(0.33)	-0.00	(0.34)	-0.00	(0.33)	0.00	(0.38)
SR_m	0.14	(0.06)	0.14	(0.11)	0.14	(0.06)	0.08	(0.93)	-0.03	(2.91)
$corr(pd, \bar{v})$	-0.57	(0.07)	-0.76	(2.45)	-0.77	(2.70)	—	—	-0.78	(2.73)
$corr(r^f, \bar{v})$	0.26	(0.11)	0.51	(2.35)	0.51	(2.33)	—	—	0.71	(4.18)
$corr(pd, r^f)$	0.08	(0.15)	-0.20	(1.96)	-0.14	(1.57)	1.00	(6.28)	-0.33	(2.86)
$\mu(pc)$	—	—	4.54	—	4.87	(1.85)	5.21	(5.59)	4.39	(3.49)
$\sigma(pc)$	—	—	0.25	—	0.14	(5.56)	0.00	(8.17)	0.13	(5.78)
$ac_1(pc)$	—	—	0.96	—	0.95	(1.07)	0.76	(14.17)	0.95	(1.10)
$\mu(r^c)$	—	—	1.54	—	1.21	(0.81)	0.98	(1.29)	1.68	(0.19)
$\sigma(r^c)$	—	—	7.13	—	4.87	(3.64)	0.98	(6.15)	4.55	(3.85)
$ac_1(r^c)$	—	—	0.02	—	0.03	(0.74)	0.44	(5.33)	0.03	(0.76)
$\mu(r^c - r^f)$	—	—	1.38	—	1.04	(0.81)	0.19	(2.54)	-0.05	(3.04)
$\sigma(r^c - r^f)$	—	—	7.06	—	4.78	(3.69)	0.77	(6.27)	4.44	(3.91)
$ac_1(r^c - r^f)$	—	—	-0.00	—	-0.00	(0.33)	-0.00	(0.34)	0.00	(0.42)

Table 4

Model configurations – quarterly frequency. Panel A shows the common preference parameters whenever they apply. They were chosen to fit the average risk-free rate, the volatility of the risk-free rate, the stock market Sharpe ratio and the persistence of the stock market log price-to-dividend in the full model. Panel B shows where the models differ. Model 1 is the full model with disagreement, stochastic information quality and stochastic risk aversion. The rest of the models remove one or more of these features. Constant information quality is imposed by setting $\sigma_p = \beta_x = \sigma_x = 0$, constant risk aversion is imposed by setting $\varphi_h = 1$, and no disagreement is imposed by setting $\eta = 0$. The subjective discount factor in each of the Models 3 to 6 was chosen to fit the average stock market log price-to-dividend ratio.

A. Common preference parameters						
Description	Parameter	Value				
Subjective discount factor	β	0.982				
Utility curvature parameter	γ	1.17				
Conditional volatility of ω at steady state	θ	23				
Degree of risk aversion at the average state (at $\omega = 0$)	$\gamma \cdot (1 + \theta)$	26.91				
Log consumption surplus ratio persistency	ϕ_h	0.985				
B. Model differences						
Description	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Subjective discount factor	0.9820	0.9820	0.9683	0.9731	0.9687	0.9633
Disagreement	yes	yes	yes	yes	no	no
Stochastic risk aversion	yes	no	no	yes	yes	no
Stochastic information quality	yes	yes	no	no	yes	yes

5.4 Tables 5–6 and Figure 3: removing disagreement and inspecting state variables

Read: Table 5:

- Model 4 keeps habit but fixes information quality.
- Model 5 removes disagreement but keeps stochastic information quality and stochastic risk aversion.
- Model 6 removes disagreement and keeps constant risk aversion.
- Removing disagreement sharply reduces the volatility generated by stochastic information quality.

Read: Table 6:

- The state variables are highly persistent.
- Forecast dispersion is strongly negatively correlated with the information-quality state and with stock valuation.
- The table documents why forecast dispersion is a useful empirical proxy for the model’s latent information-quality channel.

Read: Figure 3:

- The price-dividend ratio is most sensitive to the information-quality state χ and the habit/risk-aversion state ω .
- The risk-free rate is sensitive to market expectation, uncertainty, and forecast dispersion.
- Forecast dispersion is important for rates even if it is not itself the main direct stock-price state variable.

Economic message: The model’s central channel is visible here: information quality moves both valuation and rates because it changes disagreement and risk aversion states.

Table 5

Asset pricing statistics – Data and models with and without disagreement. The table shows a number of asset pricing moments for the data and four different model configurations. Relative to Model 1, Model 4 has constant information quality, Model 5 has no disagreement, and Model 6 has no disagreement and constant risk aversion—see parameters in Tables 2 and 4. “Habit” imposes no disagreement ($\eta = 0$) to Model 4, where we set $\beta = 0.9677$ to match the average risk-free rate of Model 4. For a variable x , $\mu(x)$ denotes the sample mean, $\sigma(x)$ the standard deviation, $ac_k(x)$ its k -lag autocorrelation and $corr(x, y)$ the correlation with variable y . The notation is as follows: pd denotes the log price-to-dividend ratio for the stock market dividend claim, r^f the log risk-free rate, r^m the log stock market return, SR_m the stock market Sharpe ratio using log returns, g^d is the log growth rate of the stock market dividend claim, pc the log price-to-consumption ratio of the per-capita consumption claim, and r^c the log return of the consumption claim. The standard errors for the data estimates $s.e.$ are computed using the Newey and West (1987) method with 16 lags. For each model we report the averages of 1,000 simulations of 400 quarters each (with a burn-in of 100 quarters). The t -statistics for the models $t-st$ refer to the hypotheses that the data estimates are equal to the model averages.

	Data		Model 4		Model 5		Model 6		Habit	
	Data	s.e.	avg.	t-st	avg.	t-st	avg.	t-st	avg.	t-st
$\mu(pd)$	4.71	(0.09)	4.71	(0.08)	4.71	(0.08)	4.70	(0.01)	4.62	(0.90)
$\sigma(pd)$	0.44	(0.05)	0.14	(5.60)	0.11	(6.14)	0.03	(7.73)	0.14	(5.60)
$ac_1(pd)$	0.97	(0.01)	0.95	(1.24)	0.95	(1.60)	0.87	(6.56)	0.95	(1.33)
$\mu(r^f)$	0.14	(0.13)	0.30	(1.16)	0.41	(2.04)	0.98	(6.22)	0.30	(1.17)
$\sigma(r^f)$	0.84	(0.09)	0.61	(2.44)	0.79	(0.54)	0.79	(0.52)	0.57	(2.92)
$ac_1(r^f)$	0.65	(0.05)	0.76	(2.24)	0.76	(2.20)	0.76	(2.21)	0.76	(2.22)
$\mu(r^m)$	1.59	(0.47)	1.34	(0.52)	1.34	(0.53)	1.34	(0.53)	1.43	(0.34)
$\sigma(r^m)$	10.52	(1.55)	8.18	(1.51)	7.66	(1.84)	6.28	(2.73)	8.27	(1.45)
$ac_1(r^m)$	-0.03	(0.09)	0.02	(0.64)	0.02	(0.65)	0.03	(0.69)	0.02	(0.62)
$\mu(r^m - r^f)$	1.44	(0.49)	1.05	(0.81)	0.92	(1.05)	0.36	(2.19)	1.13	(0.64)
$\sigma(r^m - r^f)$	10.53	(1.55)	8.17	(1.51)	7.66	(1.85)	6.25	(2.75)	8.27	(1.45)
$ac_1(r^m - r^f)$	-0.03	(0.09)	-0.00	(0.30)	-0.00	(0.30)	-0.00	(0.32)	-0.01	(0.28)
SR_m	0.14	(0.06)	0.13	(0.15)	0.12	(0.28)	0.06	(1.33)	0.14	(0.01)
$corr(pd, \bar{v})$	-0.57	(0.07)	—	—	—	—	—	—	—	—
$corr(r^f, \bar{v})$	0.26	(0.11)	—	—	—	—	—	—	—	—
$corr(pd, r^f)$	0.08	(0.15)	0.47	(2.66)	0.64	(3.81)	0.13	(0.30)	0.53	(3.05)
$\mu(pc)$	—	—	5.92	—	7.42	(30.27)	5.42	(7.92)	5.89	(13.24)
$\sigma(pc)$	—	—	0.05	—	0.03	(7.71)	0.04	(7.42)	0.04	(7.40)
$ac_1(pc)$	—	—	0.97	—	0.91	(3.77)	0.94	(2.23)	0.98	(0.50)
$\mu(r^c)$	—	—	0.70	—	0.49	(2.33)	0.88	(1.51)	0.71	(1.87)
$\sigma(r^c)$	—	—	1.87	—	1.02	(6.13)	1.02	(6.13)	1.82	(5.61)
$ac_1(r^c)$	—	—	0.18	—	0.27	(3.39)	0.17	(2.28)	0.20	(2.61)
$\mu(r^c - r^f)$	—	—	0.40	—	0.08	(2.76)	-0.10	(3.12)	0.41	(2.09)
$\sigma(r^c - r^f)$	—	—	1.78	—	0.74	(6.29)	0.69	(6.33)	1.75	(5.65)

Table 5: models with and without disagreement

Table 6

State variable moments – Model 1. The table shows the correlations, means, standard deviations (stdev), and first-lag autocorrelations (ac_1) of the state variables, for Model 1. They are computed as the averages across 1,000 simulations of 400 quarters each (with a burn-in of 100 quarters).

Correlations								mean	stdev	ac_1
$\bar{\mu}$	$\bar{v}$	v	χ	ω	$\bar{\mu}$	$\bar{v}$	ξ			
$\bar{\mu}$	1.00	-0.23	0.06	0.36	0.27	0.98	-0.23	-0.24	0.435	0.557
$\bar{v}$		1.00	-0.18	-0.73	-0.67	-0.20	0.99	0.95	0.272	0.151
v			1.00	0.06	0.13	0.02	-0.20	-0.42	0.358	0.085
χ				1.00	0.90	0.30	-0.75	-0.69	-0.368	0.954
ω					1.00	0.22	-0.70	-0.66	0.000	0.015
$\bar{\mu}$						1.00	-0.20	-0.19	0.439	0.515
$\bar{v}$							1.00	0.95	0.291	0.163
ξ								1.00	0.149	0.145

Table 6: state variable moments

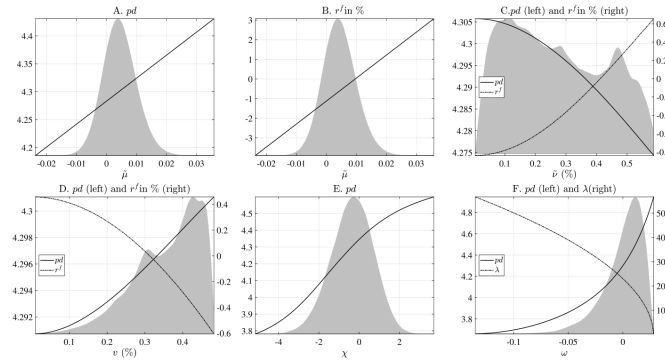


Figure 3: price-dividend ratio and risk-free rate in Model 1

5.5 Tables 7–9: short rates, real yields, and bond returns

Read: Table 7:

- Forecast dispersion predicts future cumulative short rates with positive coefficients.
- The model generates similar horizon patterns.
- The speculative activity term has the opposite sign because it is inversely related to the short-rate effect.

Read: Table 8:

- Models without stochastic disagreement have weak or negative correlations with real yields.

- Model 1 performs better, especially at long maturities.
- The model generates an upward-sloping real yield curve when stochastic disagreement is active.

Read: Table 9:

- Model 1 generates real bond return volatilities close to the data.
- The model produces positive real bond premia, unlike models with constant information quality.
- Bond return moments reinforce the same mechanism as the real yield results.

Economic message: Forecast dispersion is a state variable for the real term structure. The evidence in these tables connects belief dispersion to future short rates, real yields, and bond premia.

Table 7

Predicting short-term rates. This table reports the β coefficients of the univariate predictive regressions

$$r_{t+T}^f = \alpha + \beta x_t + \varepsilon_{t+T},$$

where r_{t+T}^f denotes the cumulative log risk-free rate, that is $r_{t+T}^f = \sum_{i=1}^T r_{t+i}^f$, where r_t^f denotes the log risk-free rate between period t and $t+1$. In Panel A the predictive variable x is $1 \div (1 + \xi)$, whose values in the data is estimated based on Model 1. In Panel B the predictive variable x is $\bar{v}$ whose values in the data is given by the cross-sectional dispersion in professional forecasts. The t -statistics for the data $t - st$ are estimated using the Newey and West (1987) method with 16 lags. We report the averages of 1,000 simulations of 400 quarters each (with a burn-in of 100 quarters). The t -statistics for the models $t - st$ refer to the hypotheses that the data estimates are equal to the model averages.

A. $r_{t+T}^f = \alpha + \beta(1 + \xi_t)^{-1} + \varepsilon_{t+T}$									
	$T=1$	$T=2$	$T=4$	$T=8$	$T=12$	$T=16$	$T=20$	$T=24$	$T=28$
Data	-0.02	-0.03	-0.07	-0.15	-0.24	-0.33	-0.43	-0.51	-0.59
$t - st$	(1.71)	(1.84)	(1.84)	(1.86)	(1.93)	(2.05)	(2.16)	(2.25)	(2.39)
Model	-0.05	-0.09	-0.15	-0.26	-0.36	-0.45	-0.52	-0.58	-0.63
$st.dev.$	0.01	0.02	0.04	0.09	0.14	0.18	0.22	0.26	0.29
$t - st$	(3.80)	(3.08)	(2.03)	(1.34)	(0.96)	(0.72)	(0.49)	(0.33)	(0.15)

B. $r_{t+T}^f = \alpha + \beta \bar{v}_t + \varepsilon_{t+T}$									
	$T=1$	$T=2$	$T=4$	$T=8$	$T=12$	$T=16$	$T=20$	$T=24$	$T=28$
Data	1.40	2.78	5.70	11.84	18.47	25.28	32.31	38.52	45.21
$t - st$	(1.91)	(2.00)	(1.98)	(1.98)	(2.02)	(2.13)	(2.26)	(2.35)	(2.49)
Model	2.69	4.52	7.63	13.54	18.94	23.58	27.49	30.74	33.40
$st.dev.$	0.72	1.39	2.72	5.28	7.67	9.93	12.09	14.10	16.01
$t - st$	(1.76)	(1.26)	(0.67)	(0.28)	(0.05)	(0.14)	(0.34)	(0.47)	(0.65)

Table 8

Real yields. The table shows quarterly data and model-predicted yields to maturity of real zero-coupon bonds, for various maturities. The 1-quarter data series is constructed as the nominal risk-free rate minus the inflation expectation of the fitted AR(1) model. The rest of the data is constructed as in Gürkaynak et al. (2008) and obtained from the Federal Reserve website. The model-predicted yields are computed using the data on mean forecast ($\hat{\mu}$) and forecast dispersion ($\hat{\sigma}$) and the model-based predictions for the rest of the state variables, v , χ , and ω . Models 1 - 6 are as specified in Table 4.

A. 1968Q3 – 2016Q4												
1 quarter	Mean (%)				Standard deviation (%)				Correlation with data			
	Data	Model 1	Model 2	Model 3	Data	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3	
	1.46	-0.01	-0.01	2.95	2.96	2.89	2.89	1.95	0.09	0.09	-0.09	
	Model 4 Model 5 Model 6				Model 4 Model 5 Model 6				Model 4 Model 5 Model 6			
	0.98	3.24	5.48		1.95	1.90	1.90		-0.09	-0.12	-0.12	

B. 1999Q1 – 2016Q4												
1 quarter	Data	Model 1	Model 2	Model 3	Data	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3	
	-0.70	-1.18	-1.18	3.12	2.54	0.94	0.94	1.13	0.04	0.04	0.01	
	2 years	0.65	-0.79	-0.83	2.50	1.72	0.38	0.38	0.54	-0.01	0.02	-0.23
	5 years	1.24	-0.22	-0.21	2.17	1.57	0.35	0.35	0.25	0.38	0.38	-0.13
10 years	1.77	0.80	0.78	2.04	1.32	0.40	0.34	0.12	0.42	0.43	-0.08	
	20 years	2.12	2.25	2.01	1.97	1.06	0.36	0.24	0.06	0.45	0.46	-0.06
	Model 4 Model 5 Model 6				Model 4 Model 5 Model 6				Model 4 Model 5 Model 6			
	1.14	3.34	5.57		1.13	1.32	1.32		0.01	-0.00	-0.00	
2 years	0.59	1.52	3.65		0.57	0.89	0.85		-0.24	-0.27	-0.27	
	0.23	0.58	2.73		0.30	0.60	0.50		-0.19	-0.29	-0.28	
	10 years	-0.05	-0.25	2.09	0.19	0.43	0.30		-0.22	-0.32	-0.29	
	20 years	-0.46	-1.25	1.60	0.13	0.29	0.16		-0.29	-0.37	-0.32	

reemant model was estimated using data from 1968Q3 fact that we cannot infer the model-implied time-series o

Table 7: predicting short-term rates

Table 8: real yields

Table 9

Excess bond returns. The table shows means and standard deviations of excess quarterly log-returns for bonds and stocks. Panel A shows these moments for both nominal and real 10-year and 20-year pure discount bonds. We estimate bond returns from yields to maturities. The data on nominal and real yields is constructed as in Gürkaynak et al. (2006) and Gürkaynak et al. (2008), respectively, and obtained from the Federal Reserve website. For the nominal 1-quarter nominal risk-free rate we consider the 3-month nominal T-bill rate, obtained from CRSP. To obtain the real risk-free rate, we adjust for expected inflation implied by the fitted AR(1) process. For stock market returns we consider the CRSP value-weighted return. Data moments are shown for three different periods, the first starting on 1999Q1, the second starting on 1981Q4, the third starting on 1971Q4, and all ending on 2016Q4. Panel B shows unconditional moments of real bond and stock excess returns for each model configuration, which are the averages of 1,000 simulations of 400 quarters each (with a burn-in of 100 quarters).

A. Mean (%)								
	Data (... – 2016Q4)			Models				
	1999Q1	1981Q4	1971Q4	Model 1	Model 2	Model 3	Model 4	Model 5
Nominal 10-year	1.19	1.63	0.85					
Nominal 20-year	1.38	2.32						
Real 10-year	1.26			0.61	0.54	-0.31	-0.45	-0.85
Real 20-year	1.54			1.01	0.78	-0.31	-0.61	-1.13
Stock market	0.95	1.62	1.22	1.66	1.25	0.54	1.05	0.92

B. Standard deviation (%)								
Nominal 10-year	4.90	5.67	6.10					
Nominal 20-year	8.95	10.82						
Real 10-year	3.59			3.29	3.01	1.73	2.17	3.74
Real 20-year	5.35			5.21	3.92	1.73	2.69	4.80
Stock market	8.78	8.34	8.65	11.61	8.94	6.52	8.17	7.66

Table 9: excess bond returns

5.6 Appendix Table A.1 and Figure B.1: decomposition and incomplete markets

Read: Table A.1:

- The Campbell-Shiller decomposition attributes price-dividend volatility to dividend growth, excess returns, the risk-free rate, and future valuation ratios.
- In finite samples, much of the variation appears as expected-return variation.
- In long simulations, the model reveals a larger role for persistent risk-free-rate variation.

Read: Figure B.1:

- The figure examines the adjustment term $h - 1$ in the incomplete-market analysis.
- Deviations are small under the calibrated parameters.
- This supports the claim that complete-market pricing is a good approximation.

Economic message: The appendix checks whether the main mechanism is robust to decomposition methods and to market incompleteness. Both checks support the benchmark conclusions.

Table A.1

Campbell-Shiller variance decomposition. This table reports the Campbell-Shiller variance decomposition of the stock market log price-to-dividend ratio, pd , into fluctuations in cash-flow growth g^d , the risk-free rate r_t^f , the stock market excess return $r_{t+1}^e = r_{t+1}^m - r_t^f$ and the stock market price dividend ratio 60 quarters ahead pd_{60} . r_{t+1}^m and r_t^f denote the log stock market return and the log risk-free rate, respectively, between period t and $t + 1$. The standard errors for the data estimates *s.e.* are estimated using the Newey and West (1987) method with 16 lags. Models 1 to 6 are as in Tables 3 and 5—see parameters in Tables 2 and 4. For each model we report the averages and standard deviations (*st.dev*) of 1,000 simulations of 400 quarters each (with a burn-in of 100 quarters). We also report the results for one long simulation of 500,000 quarters. The *t*-statistics for the models *t-st* refer to the hypotheses that the data estimates are equal to the model averages.

	g^d	$-r^m$	$-r^e$	$-r^f$	pd_{60}	$\sigma(r^e)$ (%)	$\sigma(r^f)$ (%)	$\sigma(pd)$
Data	0.12	0.70	0.79	-0.09	0.16	10.53	0.84	0.44
<i>s.e.</i>	(0.08)	(0.12)	(0.21)	(0.12)	(0.06)	(1.55)	(0.09)	(0.05)
Model 1 (<i>avg.</i>)	-0.10	1.09	0.91	0.19	0.02	11.61	0.85	0.31
<i>st.dev</i>	0.37	0.43	0.43	0.11	0.12	0.54	0.07	0.06
<i>t-st</i>	(2.74)	(3.25)	(0.57)	(2.39)	(2.26)	(0.70)	(0.10)	(2.38)
Long simulation	0.05	0.86	0.63	0.23	0.13	11.63	0.86	0.36
<i>t-st</i>	(0.92)	(1.33)	(0.73)	(2.70)	(0.52)	(0.71)	(0.21)	(1.55)
Model 2 (<i>avg.</i>)	-0.11	1.13	0.71	0.42	-0.01	8.94	0.85	0.16
<i>st.dev</i>	0.73	0.78	0.80	0.24	0.12	0.40	0.07	0.03
<i>t-st</i>	(2.80)	(3.49)	(0.37)	(4.38)	(2.74)	(1.02)	(0.11)	(5.33)
Long simulation	0.15	0.80	0.29	0.51	0.06	8.94	0.86	0.17
<i>t-st</i>	(0.33)	(0.84)	(2.40)	(5.18)	(1.56)	(1.02)	(0.21)	(5.11)
Model 3 (<i>avg.</i>)	1.45	-0.43	0.49	-0.93	-0.01	6.52	0.61	0.02
<i>st.dev</i>	2.60	2.61	2.35	0.54	0.06	0.23	0.04	0.00
<i>t-st</i>	(11.56)	(7.73)	(1.33)	(7.16)	(2.68)	(2.58)	(2.44)	(7.83)
Long simulation	2.18	-1.18	0.01	-1.19	0.00	6.53	0.62	0.02
<i>t-st</i>	(25.90)	(15.52)	(3.74)	(9.51)	(2.50)	(2.57)	(2.37)	(7.83)
Model 4 (<i>avg.</i>)	-0.31	1.27	1.23	0.04	0.05	8.17	0.61	0.14
<i>st.dev</i>	0.94	1.02	0.92	0.20	0.15	0.31	0.04	0.03
<i>t-st</i>	(5.15)	(4.52)	(2.10)	(1.09)	(1.72)	(1.51)	(2.44)	(5.60)
Long simulation	0.12	0.69	0.76	-0.07	0.21	8.19	0.62	0.17
<i>t-st</i>	(0.07)	(0.10)	(0.15)	(0.16)	(0.68)	(1.50)	(2.37)	(5.14)
Model 5 (<i>avg.</i>)	-0.40	1.34	1.56	-0.21	0.06	7.66	0.79	0.11
<i>st.dev</i>	1.19	1.26	1.08	0.38	0.16	0.31	0.07	0.03
<i>t-st</i>	(5.90)	(5.06)	(3.67)	(1.09)	(1.65)	(1.85)	(0.54)	(6.14)
Long simulation	0.16	0.63	1.07	-0.44	0.23	7.67	0.81	0.13
<i>t-st</i>	(0.43)	(0.60)	(1.35)	(3.05)	(1.09)	(1.84)	(0.30)	(5.72)
Model 6 (<i>avg.</i>)	1.61	-0.59	-0.82	0.22	-0.02	6.25	0.79	0.03
<i>st.dev</i>	3.69	3.66	3.04	1.42	0.10	0.23	0.07	0.01
<i>t-st</i>	(10.54)	(7.70)	(7.01)	(2.50)	(2.79)	(2.75)	(0.52)	(7.73)
Long simulation	0.56	0.40	-0.60	1.00	0.04	6.26	0.82	0.03
<i>t-st</i>	(5.46)	(2.45)	(6.69)	(9.41)	(1.83)	(2.74)	(0.27)	(7.69)

Appendix Table A.1: variance decomposition

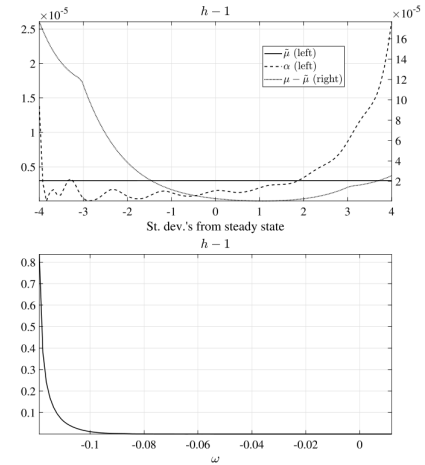


Fig. B.1. Quantity $h - 1$ in complete markets. The figure shows deviations of h from 1, where h is defined in Proposition 5, in the complete markets case of habit-formation preferences with stochastic risk aversion and constant information quality (Model 4). For these plots, we vary one state variable at a time, keeping the rest at their steady state values. For these computations we use $\bar{s} = \ln(1 - 0.93)$.

Appendix Figure B.1: incomplete-market adjustment

6 Short summary

- The paper studies asset pricing in a pure exchange economy with a continuum of agents who agree to disagree.
- The latent state is the mean growth rate of aggregate consumption.
- Agents learn from realized growth, a common signal, and individual signals.
- Information quality determines how informative the common signal is.
- Disagreement is inversely related to information quality: low public information quality creates high disagreement.
- The disagreement process is estimated with professional GDP forecasts.
- External habit preferences are used to generate a realistic market price of risk.
- The continuum of types produces a stationary equilibrium, unlike finite-agent heterogeneous-belief models.
- Constant disagreement has negligible effects on stock price volatility.
- Stochastic information quality without disagreement also has negligible effects on stock price volatility.
- The interaction of stochastic information quality and disagreement is quantitatively powerful.
- The benchmark model produces substantial price-dividend volatility and a realistic stock Sharpe ratio.
- Forecast dispersion is negatively related to stock valuations in the model and data.
- Forecast dispersion also predicts future short rates.
- Stochastic disagreement helps explain an upward-sloping real yield curve.
- Model 1 generates real bond return volatilities close to the TIPS-based data.
- Incomplete-market extensions do not overturn the main results.

7 Conclusion

7.1 Core conclusion

Xiouros and Zapatero show that disagreement is quantitatively important only when it varies over time through information-quality shocks. Constant disagreement does not generate enough stock volatility or equity premium. Time-varying information quality without disagreement is also too weak. The interaction of the two forces generates persistent movements in forecast dispersion, risk-free rates, stock valuations, and real term premia.

7.2 Economic intuition

When public information is precise, investors' beliefs are anchored by a common signal. When public information becomes less precise, investors put more weight on their own interpretations. This raises forecast dispersion. Since pessimistic and optimistic beliefs have asymmetric effects on state prices, higher forecast dispersion changes the risk-free rate. Because information quality is linked to aggregate shocks, the resulting volatility is systematic and priced.

7.3 Takeaway

The paper's main lesson is that belief heterogeneity should not be modeled only as a fixed cross-sectional disagreement parameter. The relevant pricing object is the state-dependent quality of information that

determines how dispersed beliefs become. This mechanism helps explain stock volatility, the equity premium, forecast-dispersion patterns, and the real yield curve in a single stationary general-equilibrium framework.